

We make the world more Clean and Safety

ZERONA Air Purification Sterilizer

2026. 01



We make the world more Clean and Safety

ZERONA

Under the philosophy of
"Making spaces comfortable and people safe,"
ZERONA continuously researches and develops
advanced technologies to improve indoor air quality,
prevent respiratory diseases in modern society,
and contribute to human health and happiness.
We aim to create

"A world where breathing is easy."

Contents



Core Technology



Performance &
Certification



Product
Introduction

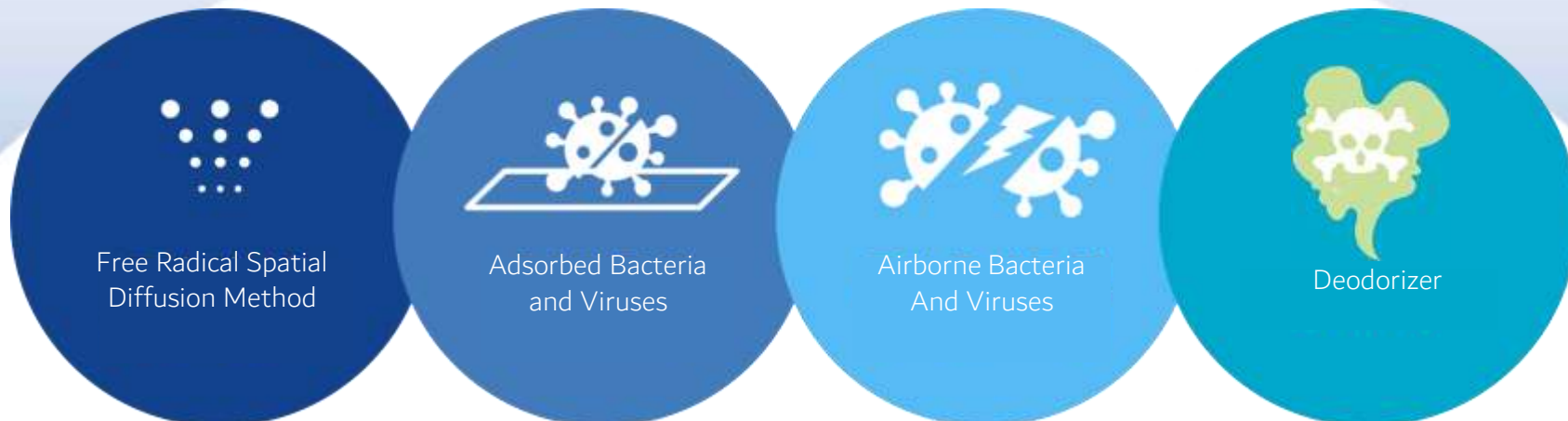
A close-up photograph of a person's hands holding a small, square microchip. The chip is held between the thumb and index finger of the left hand, positioned directly above a blue printed circuit board (PCB). The PCB is densely packed with various electronic components, including capacitors, resistors, and integrated circuits. In the background, another person's hand is visible, working on a similar circuit board. The entire scene is overlaid with a semi-transparent dark blue filter, and the text "Core Technology" is centered in white.

Core Technology

Air Purification + Sterilization System

This innovative system sterilizes and maintains clean indoor environments by eliminating:

- Airborne and surface-attached viruses such as COVID-19, RSV, Influenza, Norovirus, Rotavirus
- Harmful bacteria including E. coli, Salmonella, Staphylococcus, Tuberculosis bacteria using a **free-radical spatial diffusion method**.



Sterilization Principle

Highly reactive **free radicals** are released into the air, neutralizing or decomposing harmful substances such as:



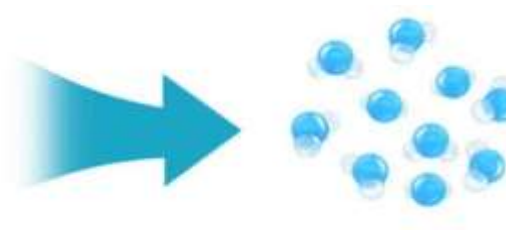
Radical Generator



Free-Radicals



Hazardous Substances



Air Sterilization, Deodorization

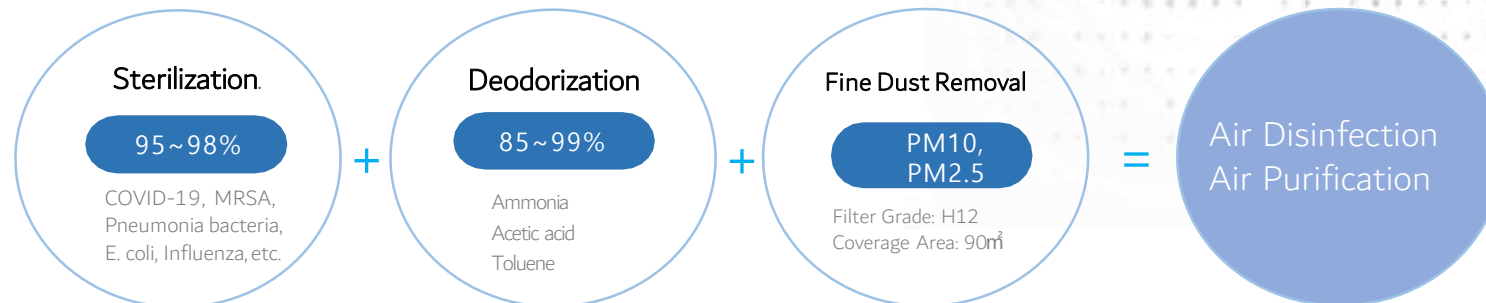




Performance & Certification

The background image features a hand holding a pen, pointing towards a large, faint gear. A network of circular icons is overlaid on the image, connected by lines. The icons include: a brain with a Wi-Fi signal, a computer monitor with a double-headed arrow, a factory, a magnifying glass over a gear, a checkmark inside a square, a bar chart with an upward arrow, and a flowchart with arrows. The overall theme is technological performance and certification.

Performance & Certification



2-meter distance test

MICROBAC 1 2m거리 살균력 시험 (2020,2021)

Fetal bovine serum (FBS) 5.0%

소태아혈청 5.0% (FBS)

Test Conditions: Organic Soil Load: 5.0 x 10⁸ CFU/g (FBS)
 Contact Times: 1 hour, 2 hours, 4 hours
 Contact Temperature: 21°C
 Contact Relative Humidity: 45-48%

COVID-19 Virus

Challenge Virus: Severe Acute Respiratory Syndrome-Related Coronavirus 2 (SARS-CoV-2) (C-19-2019-01)
 • Strain: USA-WA1/2020
 • Source: BEI Resources, NR-52281

Test Substance:

The test device was placed 1-2 m away from the prepared test carriers.

종류	종목	노출시간	복제번호	초기바이러스함유량 (Log ₁₀ TCID ₅₀)	Log ₁₀ 감소	감소퍼센트
COVID-19 (2020.09)	4 hours	1		4.73	1.17	93.19%
		2		4.48	1.55	97.16%
		3		4.6	1.55	97.16%
		Mean Reduction		1.42	96.20%	
COVID-19 (2021.07)	4 hours	1		5.73	0.75	82.35%
		2		5.85	1.5	96.86%
		3		5.98	1.5	96.86%
		Mean Reduction		1.25	94.42%	
영국변이바이러스 (2021.07)	2 hours	1		5.73	1.00	88.95%
		2		5.73	0.88	85.43%
		3		5.6	0.5	74.09%
		Mean Reduction		0.79	83.90%	

GLP 국제인증기관인 Microbac Laboratories, Inc.에서 진행된 살균력 측정시험에서는 5%의 소태아혈청(유기물)을 함유한 조건에서 진행했다는 것을 확인할 수 있습니다
 *소태아혈청은 세포 배양에 사용되는 다양한 단백질로 이루어진 유기물입니다

KTL 부유세균 및 바이러스 저감시험 (2021)

KTL Airborne Bacteria and Virus Reduction Test

1. 시험결과

종 명	결 과
AURAVAC 공간방역기	부유세균 저감율 94.2 %

3. 시험방법 및 조건

시험방법	부유세균	온도	습도	시험 챔버	시험 시간	운전모드	운전종말
KOUVA AS 02 : 2019, 공기정화장치	<i>S. enteritidis</i> (ATCC 12228)	(23 ± 2) °C	(50 ± 5) % RH	60 m³	1 시간	단일	최대 (3단)

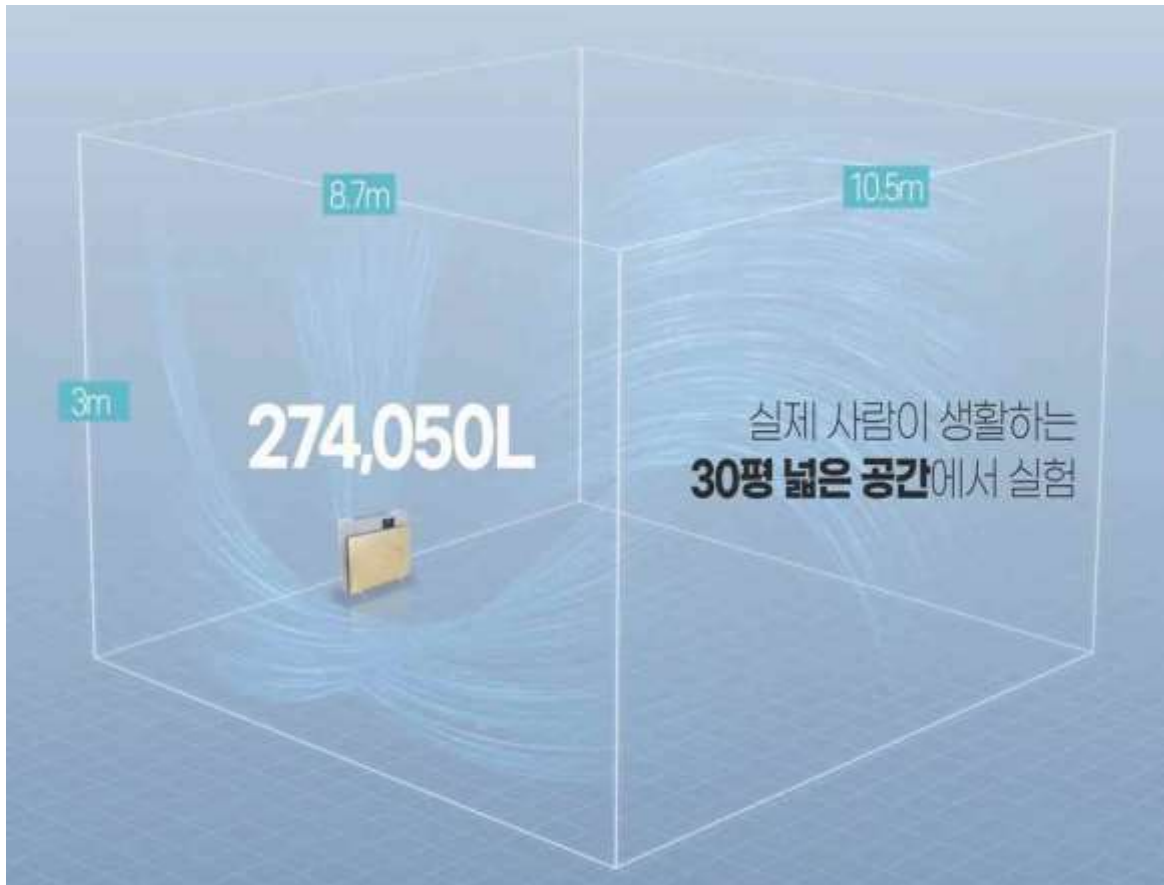
1. 시험결과

종 명	결 과
AURAVAC 공간방역기	부유바이러스 저감율 75.0 %

3. 시험방법 및 조건

시험방법	바이러스	온도	습도	시험 챔버	시험시간	운전모드	운전종말
KOUVA AS 02 : 2019, 공기정화장치	Phi-X174 (ATCC 13706-B1)	(23 ± 2) °C	(50 ± 5) % RH	60 m³	30 분	단일	최대 (3단)

Proven effectiveness in approx. 99 sq. ft. space



Konkuk University 10-meter Disinfection Range Test (2021)

건국대학교 10m 거리 살균력 실험(2021)

○ 공간 조건 : 10.5m x 8.7m x 3m, 274m³, 274,050L

시험 성적서

건국대학교 산학협력단 주 05029 서울특별시 광진구 능동로 120 (Tel : 02-450-6055 / E-mail : ischo@konkuk.ac.kr)		성적서 번호 : 2021-0510-002 페이지 (1 / 총 1)
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1. 의뢰자
 ○ 기 관 명 : 디에스브이(DSV Co., Ltd)
 ○ 주 소 : 경기도 광주시 도척로 699번길 30-99
 ○ 의뢰일자 : 2021년 4월 1일

2. 시험실적서의 용도 : 제품 품질 관리

3. 시험 시료명/품목 : 디에스브이가 제조하는 공기 살균 장치, 이온화장치 'DSV-1000'

4. 시험 기간 : 2021년 4월 1일 ~ 2021년 6월 30일

5. 시험 방법 : 바이러스 및 세균에 대한 표면 살균력 테스트 (Virucidal / bactericidal Efficacy Hard-surface carrier test)

세부 시험 방법
 1) 병원체를 10⁷ TCID₅₀/ml 배양 후 용기물 (소대아혈청, PBS) 5% 첨가 후 정사각형 모양의 Polypropylene (PP) 부른 형태에 분주 후 기기 노출 후 샘플 수거 진행.
 2) DSV-1000에 노출되지 않은 병원체 정사각형 Polypropylene (PP) 부른 용기물로 설정하여 음성대조군 대비 실험군의 병원체 역가 감소율을 비교함.

6. 시험 환경
 ○ 온도 조건 : (25 ± 3) °C, 습도 : (30 ± 3) % 및 36 ± 2 %
 ○ 공간 조건 : 8.7m x 2.8m x 2.7m, 65.772m³, 65.772L

7. 실험 진행 병원체 : Coronavirus (Covid-19 surrogate: Canine Corona virus)

8. 실험 결과

표1. 시험 1 시간 및 거리별 바이러스 역가 감소율

	2h	6h	10h
8m	82.28%	89.20%	94.38%

표2. 시험 2 후 시간 기기 작동 이후 시간 및 거리별 바이러스 역가 감소율

Hourly rate of bacterial titer reduction

	2h		2h
8m	82.28%	10m	81.571%

Safety Evaluation (1)

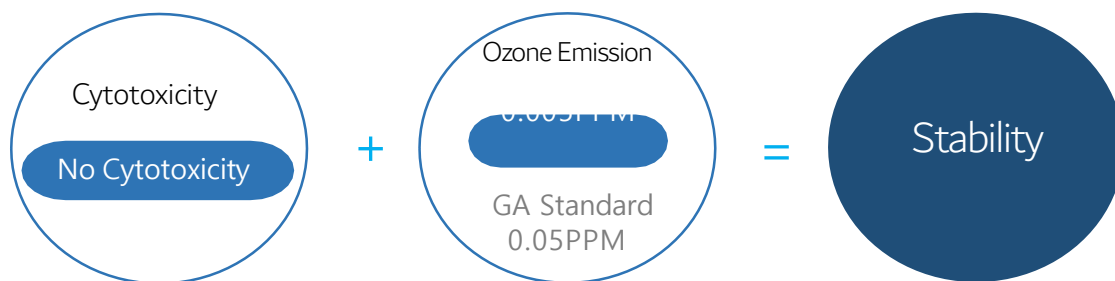


Table 2
Cytotoxicity Controls

Dilution of the Neutralized Sample	Cytotoxicity Control
10^{-1}	<u>no cytotoxicity</u> observed in 4 out of 4 wells
10^{-2}	<u>no cytotoxicity</u> observed in 4 out of 4 wells
10^{-3}	<u>no cytotoxicity</u> observed in 4 out of 4 wells

세포독성시험 결과: 세포독성 없음

Certified by Microbac Laboratories, Inc.

Cytotoxicity testing confirmed the absence of toxicity

 한국시험기술연구원 Korea Testing Laboratory	연락처 번호 : 22-663624-01-3 Report No. 페이지 (2 / 총 2) Page of Page	
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시 험 결 과 (Test Results)

1. 시험결과

종 명	기준	오존발생농도 시험결과 (과대값)
		g/mol/mol
불가탈환기	0.05 g/mol/mol 이하	0.005

2. 시험설비 및 장비

시험항목	시험장비		
	장비명	제조사	모델명
30 m ³	오존측정기	Teledyne APL Inc	400E

3. 시험방법 및 조건

시험방법	시험조건					
	온도	습도	측정거리	시험시간	준전도도	준전용량
SPS-KOUVA AS-01-1889-202, 포기함공기	(23 ± 2) ℃	(50 ± 5)% RH	5 cm	8 시간	단일	단일

4. 시험종 사전



Ozone Stability Certification

Safety Evaluation (2)

WOJUNGBIO

Final Report

Acute Inhalation Toxicity Test of the Radicle
(ZERONA, DB-12885B12L) Using a Sprague-Dawley Rat

Test Institution	Woojung Bio Co., Ltd.
Test Number	SH124004
Date of Report	September 04, 2024

5. Conclusion and Remarks

In this study, the acute inhalation toxicity of a radicle generated by the space disinfection device (ZERONA, DB-12885B12L) was evaluated by exposing experimental animals to the radicle for 4 hours per day for 1 day.

The test substance was generated by operating the radicle in a separately manufactured acrylic chamber, and the generated radicle was introduced into the portable chamber and delivered to the experimental animals through a pump. The exhaust gas was circulated in a closed loop and transferred again to the acrylic chamber to ensure that the generated radicle was maintained at maximum levels.

After inhalation exposure to the radicle, no animals showed mortality or any particular clinical signs. There were no statistically significant differences in body weight between the test and control groups. At necropsy, a reddish discoloration was observed in one animal (No. 2204) in the test group; however, the degree was mild and no such findings were observed in other animals, so it was determined to be an incidental finding.

Based on the results of this study, no adverse effects on body weight, general symptoms, or necropsy findings were observed when the radicle generated by the space disinfection device was inhaled for 4 hours. Therefore, it is concluded that the radicle has no acute inhalation toxicity.

First in the industry to conduct: Cytotoxicity test
Inhalation toxicity test
Result: Verified harmless to the human body

Deodorization Test Report

 한국산업기술시험원 Korea Testing Laboratory	실적서 번호 : 21-026700-01-I Report No.	
	페이지 (2) (총 2) Page of Pages	

시험결과 (Test Results)

1. 시험결과

품 명	유해물질 탈취율 (%)		
공기 살균기	암모니아	톨루엔	아세트산
	85.5	99.9 이상	99.15

2. 시험설비 및 장비

품 명	유해물질 탈취율 (%)		
공기 살균기	Ammonia 암모니아	Toluene 톨루엔	Acetic acid 아세트산
	85.5	99.9 이상	99.15

KOUVA AS 01 : 2016 공기살균기	(23 ± 2) °C	(50 ± 5) % RH	60 분	최대
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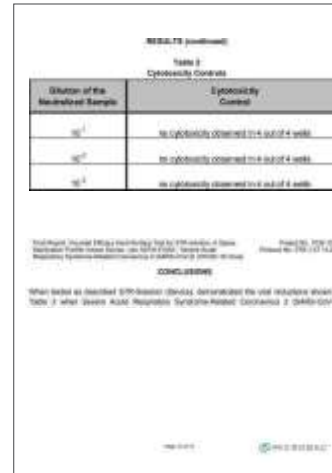
Certificate



GA Certification



MICROBAC COVID-19



MICROBAC Cytotoxicity Test



Airborne Bacteria Reduction



Airborne Virus Reduction



KTR Ozone & Noise



KTL Deodorization



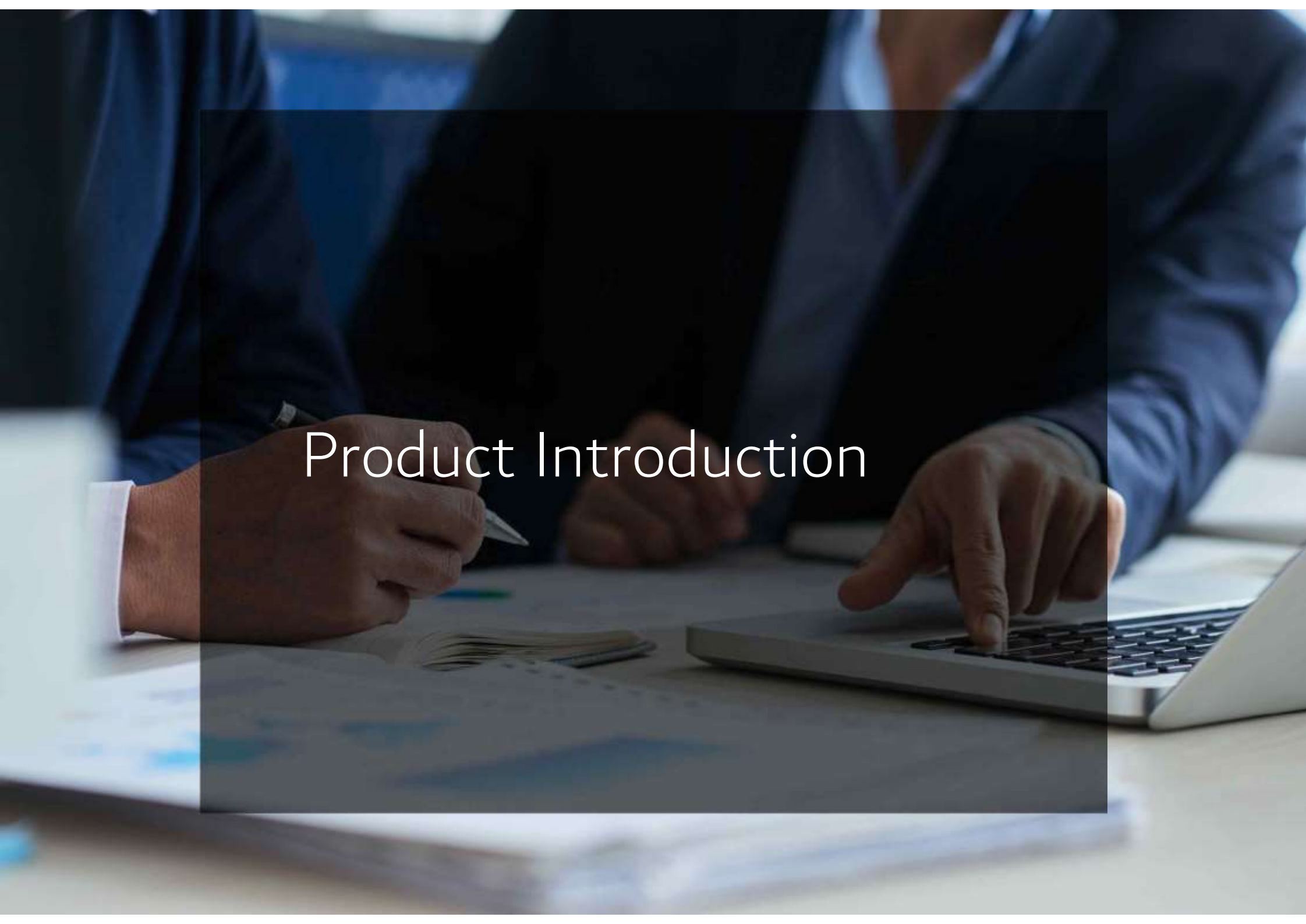
Staphylococcus



EMC Compliance



KC Electrical Safety

A blurred background image of a business meeting. Two people in dark blue suits are seated at a table. The person on the left is holding a pen over a document with charts. The person on the right is pointing at a laptop keyboard. The scene is dimly lit, with a focus on the hands and the work materials.

Product Introduction

ZERONA Air Purification Sterilizer99m³

TFT Touch Screen

3-step airflow control

- High: Use within 5 minutes after ventilation
- Medium: Use in crowded environments
- Low: General use
- Indications for filter and lamp intervals

Filters

Electrostatic filter, deodorizing filter

- Lasts 3 to 6 months, depending on usage conditions
- Can be replaced after simple disassembly

Generator

5-layer photocatalytic module

- Free radicals are generated
- Diffuse into the surrounding space
- Lamp is replaced every 24 months

Intake fan

Inhalation of polluted air & Emission

- Filtering contaminated outside air
- Delivering the filtered air to the generator

ZERON

Removes both airborne and surface-attached bacteria and viruses across the entire space.

1

Contaminated air is drawn into the device

2

Air passes through sterilization module

3

Sterilized air and free radicals are released

4

Free radicals sterilize airborne and surface contaminants

Specification

Model	SV-1010
Size	500(W) X 140(D) X 405(H) mm
Weight	9.8kg
Power	12W (AC110~240V, 12V 3.5A)
Installations	Stand Type
Filters	HEPA(H13) ,Deodorization Filter
Sensors	Temp+Humidity+PM2.5+PM10
KC Cert	Complete

Core Technology

Performance & Certification

Product Introduction

ZERONA Air Purification Sterilizer

66m³

Sterilization method

Photocatalytic (TiO₂) air diffusion system, HEPA filter



Purifies airborne viruses and surface germs!

Effectively removes harmful airborne viruses and bacteria, including 99.99% of airborne viruses.



Proven disinfecting performance against COVID-19

Certified by the CDC, a U.S. institution, for its COVID-19 disinfecting performance in actual use.



365 days, 24/7 hassle-free space care

Provides an optimal solution for harmful substances in the air such as fine dust, allergens, and odors.



Removes germs, odors, and harmful substances all at once

Effectively removes VOCs and various odors to keep indoor air fresh and clean.



Domestic & International Tests (including KOTITI, Shinheung, FITI, etc.)

Proven disinfecting performance against viruses!

99.99% Certified!

Color | ○ WHITE ● BLACK
Dimensions | 380x110x100(mm)
Weight | 1.9kg
Filter | HEPA Filter & NO Filter
Power | AC Adapter (100~240V AC)
Installation | Stand / Wall-mounted

Rental system

01 Systematic Air Quality Management

02 Accommodation Industry

HDC SPACE CARE SERVICE
HYUNDAI DEVELOPMENT



Specifications

Model	ZA-020
Size	380(W) X 110(D) X 114(H) mm
Weight	1.32kg
Power	18W (AC100~220V, 12V 2A)
Installation	Stand / Attached
Filter	HEPA (H13)
Sensor	PM2.5
KC Cert	Complete

ZERONA Air Purification Sterilizer33m²

Sterilization method

Photocatalytic (TiO₂) air diffusion system, HEPA filter

- 1 Draws in contaminated air
- 2 Removes harmful substances through pre-filter and HEPA filter
- 3 Expands and removes microorganisms generated inside
- 4 Eliminates bacteria and viruses from air and surfaces



Case



Product Specifications

Model	DSV-C700
Size	320(W) X 80(D) X 95(H) mm
Weight	0.8kg
Power	8W (DC110~240V, 12V 3A)
Installation	Stand/Attached
Filters	HEPA(H13) , Deodorization filter
Sensors	Remote control, Lamp replacement
KC Cert	Complete